

Portfolio Construction with Hedge Demand

Session 13 • Merton's hedge demand • LP allocation under stochastic $L(t)$

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

Graduate Finance Course • Spring 2027 • Session 13 of 32

What we'll cover today

1

Mean-variance limitations

Why MVO fails for private capital

2

Merton's intertemporal model

Adding state variable dynamics

3

The hedge demand term

Allocation responds to $L(t)$ changes

4

GE-LAV portfolio implications

Reduced PE allocation, vintage diversification

5

Empirical: pension fund examples

How real LPs adjust

Why mean-variance optimization fails for private capital

Markowitz MVO is the bedrock of portfolio theory, but rests on assumptions PE violates:

Single-period assumption

MVO optimizes one-period return/risk. PE is a 10-year commitment.

Need: intertemporal framework

Liquid trading assumption

MVO assumes continuous rebalancing. PE has lock-up periods and J-curves.

Need: lock-up constraints

Stationary distribution

MVO assumes IID returns. PE returns are regime-dependent and autocorrelated.

Need: state-dependent dynamics

Constant risk premium

MVO assumes a fixed Sharpe ratio. PE risk premium varies with $L(t)$.

Need: stochastic premium

Merton's intertemporal portfolio model

Merton (1971): allocation between risky and risk-free assets when investment opportunities are time-varying.

$$\pi^* = \frac{\mu - r}{\gamma \sigma^2} + \frac{1}{\bar{\gamma}} \cdot \sigma_{rL} \cdot \frac{\partial V / \partial L}{\partial V / \partial W}$$

Two terms:

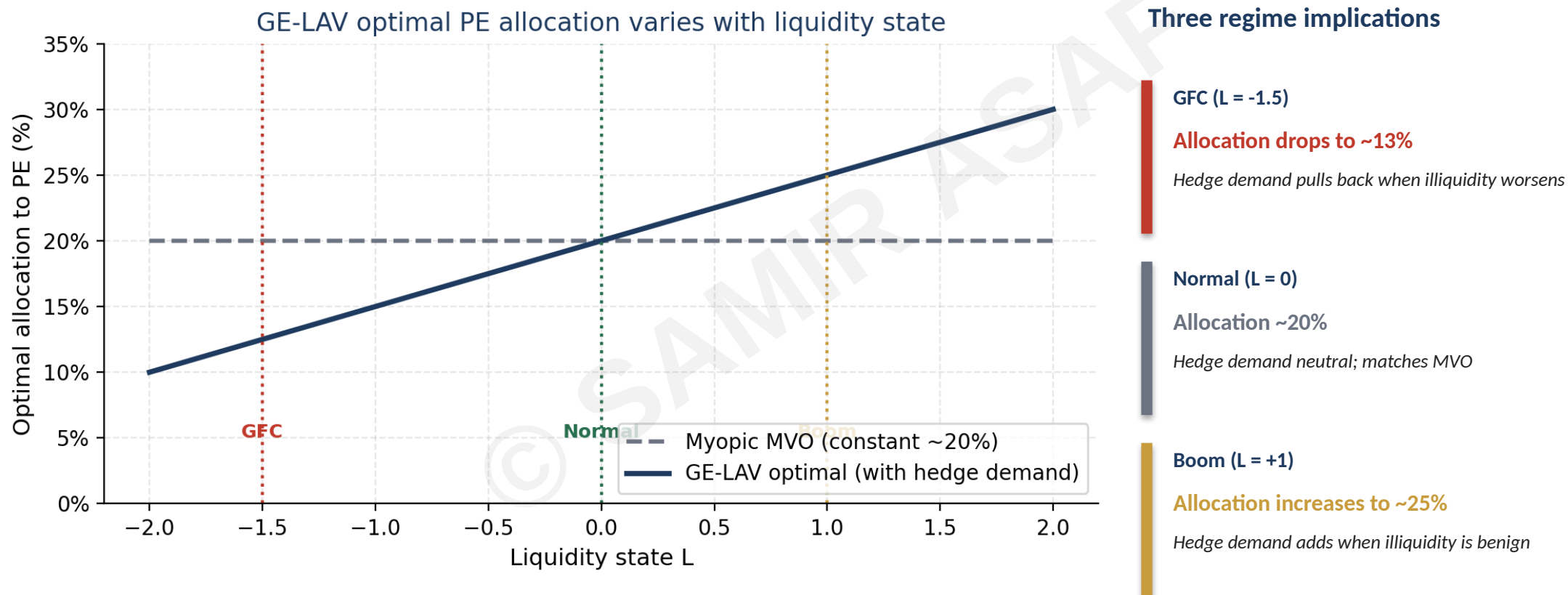
Myopic term

The standard MVO allocation: proportional to risk premium, inversely proportional to variance and risk aversion γ . Captures static risk-return tradeoff.

Hedge demand term

The Merton correction: depends on σ_{rL} (return-state covariance) and the sensitivity of indirect utility to the state. Captures dynamic hedging against state shocks.

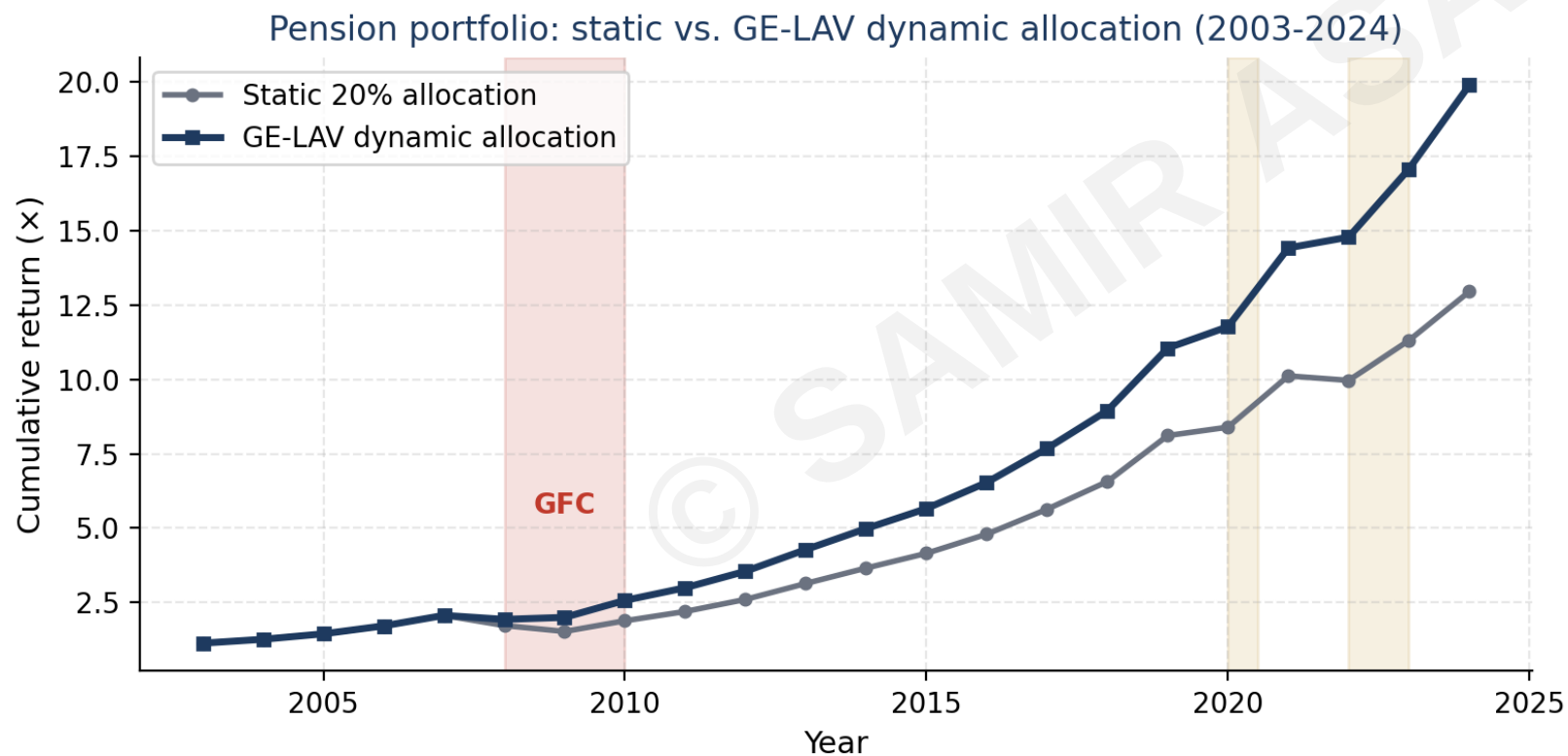
GE-LAV: how hedge demand changes PE allocation



Practical implication: dynamic allocation, not static.

Pension fund case: dynamic vs. static allocation

Hypothetical \$50B pension with 20% target to PE — comparing static and GE-LAV-dynamic strategies over 2003-2024.



Outcomes (2003-2024)

Static

8.2%/yr

Sharpe 0.62

GE-LAV

9.6%/yr

Sharpe 0.78

Improvement

+1.4%/yr

+26% Sharpe

Session 13 summary

What we accomplished today

- 1 Mean-variance optimization fails for PE due to four structural assumption violations
- 2 Merton's intertemporal framework introduces the hedge demand term — allocation responds to state changes
- 3 GE-LAV optimal PE allocation varies from ~13% (GFC) to ~25% (boom); not constant
- 4 Backtested 2003-2024: dynamic allocation adds ~1.4%/yr and 26% Sharpe improvement

Next session

Session 14: Project proposal due. Topics for guidance discussion.